

Experiment No: 7

Title: Offline/Online Migration of Virtual OS

Theory:

Migrating Virtual Machines

Migration is the process of moving a virtual machine from one host or storage location to another. Copying a virtual machine creates a new virtual machine. It is not a form of migration.

In vCenter Server, you have the following migration options:

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| Cold Migration | Moving a powered-off virtual machine to a new host. Optionally, you can relocate configuration and disk files to new storage locations. You can use cold migration to move virtual machines from one datacenter to another. |
| Migrating a Suspended Virtual Machine | Moving a suspended virtual machine to a new host. Optionally, you can relocate configuration and disk files to new storage location. You can migrate suspended virtual machines from one datacenter to another. |
| Migration with vMotion | Moving a powered-on virtual machine to a new host. Migration with vMotion allows you to move a virtual machine to a new host without any interruption in the availability of the virtual machine. You cannot use vMotion to move virtual machines from one datacenter to another. Some configurations include Metro vMotion, a feature that enables reliable migrations between hosts separated by high network round-trip latency times. Metro vMotion is automatically enabled when the appropriate license is installed. No user configuration is necessary. |
| Migration with Storage vMotion | Moving the virtual disks or configuration file of a powered-on virtual machine to a new datastore. Migration with Storage vMotion allows you to move a virtual machine's storage without any interruption in the availability of the virtual machine. |

Both migration of a suspended virtual machine and migration with vMotion are sometimes referred to as "hot migration", because they allow migration of a virtual machine without powering it off. Migration with vMotion is sometimes referred to as "live migration".

You can move virtual machines manually or set up a scheduled task to perform the cold migration.

Cloning a virtual machine or copying its disks and configuration file creates a new virtual machine. Cloning is not a form of migration.

Virtual Machine Configuration Requirements for vMotion

A number of specific virtual machine configurations can prevent migration of a virtual machine with vMotion.

The following virtual machine configurations can prevent migration with vMotion:

- You cannot use migration with vMotion to migrate virtual machines using raw disks for clustering purposes.
- You cannot use migration with vMotion to migrate a virtual machine that uses a virtual device backed by a device that is not accessible on the destination host. (For example, you cannot migrate a virtual machine with a CD drive backed by the physical CD drive on the source host.) Disconnect these devices before migrating the virtual machine. Virtual machines with USB passthrough devices can be migrated with vMotion as long as the devices are enabled for vMotion.
- You cannot use migration with vMotion to migrate a virtual machine that uses a virtual device backed by a device on the client computer. Disconnect these devices before migrating the virtual machine.

Host Configuration for vMotion

In order to successfully use vMotion, you must first configure your hosts correctly.

Ensure that you have correctly configured your hosts in each of the following areas:

- Each host must be correctly licensed for vMotion.
- Each host must meet shared storage requirements for vMotion.
- Each host must meet the networking requirements for vMotion.

Important

The ESXi firewall in ESXi 5.0 does not allow per-network filtering of vMotion traffic. Therefore, you must install rules on your external firewall to ensure that no incoming connections can be made to the vMotion socket.

Migrate a Powered-Off or Suspended Virtual Machine in the vSphere Web Client

You can use the Migration wizard to migrate a powered-off virtual machine or suspended virtual machine.

Procedure

- 1 Select a virtual machine.
 - In the virtual machines and templates inventory tree, select a group of virtual machines and select a virtual machine from the list on the right.

- Search for a virtual machine and select it from the search results list.

- 2 Right-click the virtual machine and select **Inventory > Migrate**.
- 3 Select the migration type.

Option	Description
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- 4 Select the destination resource pool for the virtual machine migration and click **Next**.
- 5 In the Host Name column, select the destination host or cluster for this virtual machine migration and click **Next**. Any compatibility problem appears in the Compatibility panel. Fix the problem, or select another host or cluster. Possible targets include hosts and DRS clusters with any level of automation. If a cluster has no DRS enabled, select a specific host in the cluster rather than selecting the cluster itself.
- 6 Select the datastore location where you want to store the virtual machine files.
- 7 If you chose to move the virtual machine's configuration file and virtual disks, select a disk format.
- 8 Review the information on the Review Selections page and click **Finish**.

vCenter Server moves the virtual machine to the new host. Event messages appear in the **Events** tab. The data displayed on the Summary tab shows the status and state throughout the migration. If errors occur during migration, the virtual machines revert to their original states and locations.

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Migrate a Powered-On Virtual Machine with vMotion in the vSphere Web Client

You can use the Migration wizard to migrate a powered-on virtual machine from one host to another using vMotion technology. To relocate the disks of a powered-on virtual machine, migrate the virtual machine using Storage vMotion.

Prerequisites

Before migrating a virtual machine with vMotion, ensure that your hosts and virtual machines meet the requirements for migration with vMotion.

- [Host Configuration for vMotion](#)
- [Virtual Machine Configuration Requirements for vMotion](#)

Procedure

- 1 Select a virtual machine.
 - In the virtual machines and templates inventory tree, select a group of virtual machines and select a virtual machine from the list on the right.
 - Search for a virtual machine and select it from the search results list.
- 2 Right-click the virtual machine and select **Inventory > Migrate**.
- 3 Select **Change host** and click **Next**.
- 4 Select the destination resource pool for the virtual machine migration and click **Next**.
- 5 Select a destination host or cluster for the virtual machine.
- 6 Select the migration priority level and click **Next**.

Reserve CPU for optimal VMotion performance

Perform with available CPU resources

- 7 Review the page and click **Finish**.

Conclusion: - Thus we have performed Offline/Online Migration of Virtual OS.